



Examination for Fundamentals of Computer Science (5DV037)

2025-03-17, 08:00-12:00

To be filled out by the student

Please print carefully and readably. Ensure that all submitted pages carry the code number X92 (as shown on the top right of this page). Submit all 9 numbered pages of the exam. Further instructions on the next page.

Name: _____

UmU-id: _____

Personnummer: _____

Today's date: _____

Submission time: _____

Prefilled information from the department

Program: Examination in computing science, contact person Ellinor Hansson.

Course title/module: 5DV037 Fundamentals of Computer Science (Datavetenskapens Grunder).

To be filled out by the exam invigilator

☐ The student has submitted a blank exam Signature: _____ Time: _____



Examination for Fundamentals of Computer Science (5DV037)

2022-03-17, 13:00-17:00

Material permitted: Pens and an English dictionary.

Number of problems and grades: This exam has a *total of 10 questions*, the requirement for grades are as follows.

- For a grade of 3 you need: At least 50% of the 50 total possible points.
- For a grade of 4 you need: At least 67% of the 50 total points, *and* at least 1 of the 3 possible points on the final three questions which are marked with a ☆.
- For a grade of 5 you need: At least 83% of the 50 total points, *and* at least 2 of the 3 possible points on the final three questions which are marked with a ☆.

Notice that the ☆-marked problems are a bit harder, but do not give very many points. Thus you may want to make them low priority unless shooting for a higher grade.

Instructions: read carefully before you begin writing your exam!

- Begin by writing your name, UmU-id, and 'personnummer' on the cover sheet. Double-check that the code number is correctly entered on every sheet.
- The space provided on the exam itself is designed to be sufficient to answer the questions, excessively long answers may have points deducted.
- If needed you may give answers on additional sheets (e.g. if you can come up with no shorter answer, or if you accidentally ruin a sheet), but make sure to write *your code number* and the *number of the question* on that sheet. Answer at most one question on each additional sheet.
- Make sure to answer questions in full whenever possible, notably where motivation is requested make sure you provide it.
- Find attached (on the last page) some helpful definitions from the course.

Question	Answered	Score
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
Total score:		
Grade:		

Code number:

Question 1 (6 points): For each statement below indicate whether it is true or false. *Note on scoring: Incorrect answers give -0.5 points, but the total for this question is never less than zero.*

Statement	True	False
The language over $\{a, b, c\}$ which contains a string w if and only if it either starts with an a , ends with a c , or contains an odd number of b s is a regular language.	☐	☐
If L is a (non-regular) context-free language and L' is a regular language then it is necessarily the case that $L \cup L'$ is <i>not</i> regular.	☐	☐
If L is a (non-regular) context-free language and L' is a regular language then it is necessarily the case that $L \cap L'$ is <i>not</i> regular.	☐	☐
If $P = NP$ then it follows that there exists a polynomial time algorithm which decides whether propositional logic formulas are satisfiable.	☐	☐
It can be correct to say that a Turing machine M is <i>decidable</i> .	☐	☐
The set of suitably encoded Turing machines is recursively enumerable.	☐	☐

Question 2 (5 points): Argue, with as much detail as space permits, that the set of regular languages is closed under complement, intersection, concatenation and union.

[illegible]

Code number:

Question 3 (8 points): Below is outlined a hierarchy of five language classes. Select five of the classes we have learned about during this course and for each fill in: (a) the name of the language class; (b) one example of a type of device/automata/grammars that can represent all the languages in that class; and; (c) one example of a language which is in that class but not in any of the subclasses you have included. The language does not have to be given as a grammar or automaton, it can be carefully described informally or using “setbuilder” notation (e.g. $\{a^n b \mid n \in \mathbb{N}\}$).

(a). Name of class

1. _____

2. _____

3. _____

4. _____

5. _____

(b). Representable by

1. _____

2. _____

3. _____

4. _____

5. _____

(c). A language from that class

1. _____

2. _____

3. _____

4. _____

5. _____

A note on grading: points are given primarily for how your answers *agree*. That is, even if a correct language class is given in (a) it will not be given points if the answer doesn't agree with the corresponding answer in (b) and/or (c).

Code number:

Question 4 (8 points): Prove that $L = \{w \in \{a, b\}^* \mid |w|_a > 2|w|_b\} \cup \{w \in \{a, b\}^* \mid 2|w|_a < |w|_b\}$ is *not* context-free. That is, L is the language of all strings which *either* contain more than twice the number of *as* then *bs*, *or* more than twice the number of *bs* than *as*.

[illegible]

Code number:

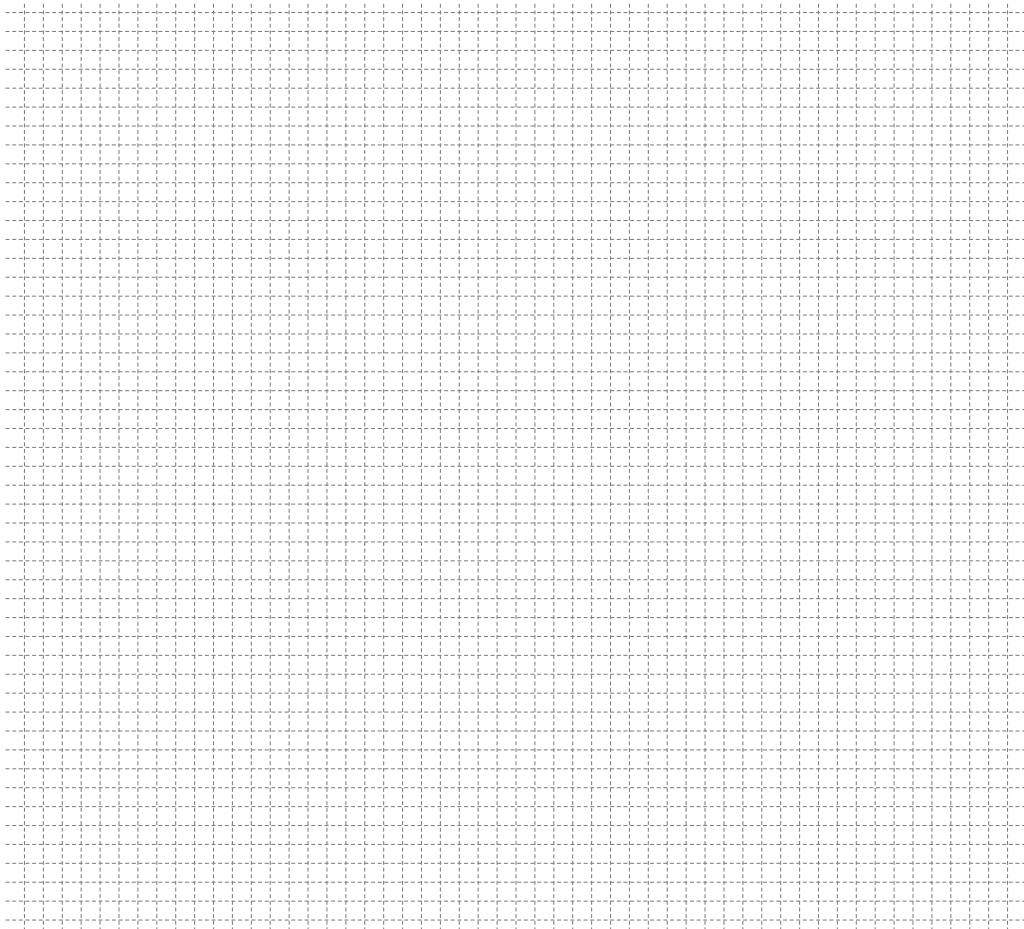
Question 5 (8 points): Consider the context-free grammar $(\{S, S', X, Y, B, C\}, \{a, b, c\}, P, S)$ with the productions in P as below.

$$\begin{aligned} S &\rightarrow SS \mid XY \mid XS' \\ S' &\rightarrow SY \\ X &\rightarrow a \\ Y &\rightarrow BC \mid BB \mid b \\ B &\rightarrow b \\ C &\rightarrow CC \mid c \end{aligned}$$

Is this grammar on Chomsky normal form?

- ☐ Yes
- ☐ No, one production breaks the normal form
- ☐ No, several productions break the normal form

Demonstrate how the string $abcab$ is parsed in this grammar using Cocke-Younger-Kasami. Show the resulting table and very briefly explain what the result means.



Code number:

Question 6 (6 points): Describe how to construct a Turing machine $M_{g \circ f}$ which computes a function $g \circ f(v)$ (i.e. started on $v \in \Sigma^*$ the machine should halt with the output $g \circ f(v)$ on its tape) given a Turing machine $M_f = (Q_f, \Sigma, \Gamma, \delta_f, q_{0,f}, \square, F_f)$ which computes the function f , and a Turing machine $M_g = (Q_g, \Sigma, \Gamma, \delta_g, q_{0,g}, \square, F_g)$ which computes the function g . Describe the elements of the tuple defining $M_{g \circ f}$ is as much detail as space permits.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 7 (6 points): Very briefly sketch why the class of nondeterministic Turing machines is no more computationally powerful than the class of deterministic Turing machines.

[illegible]

Question 8 (1 points, ☆): Define the binary operator $\diamond$ so that $(0 \diamond 0) = (0 \diamond 1) = (1 \diamond 0) = 1$, and $(1 \diamond 1) = 0$. Then the $\diamond$ -EQUATION problem is formally defined as follows.

Input: An equation $e_1 = e_2$ where e_1 and e_2 are both expressions which combine variables, the constants 0 and 1, using the operator $\diamond$.

Question: Is there an assignment of values from $\{0,1\}$ to the variables which makes the equation true?

Prove that the $\diamond$ -EQUATION problem is NP-complete with as much detail as the space provided permits.

Example of problem: Input $((x \diamond 1) \diamond y) \diamond (x \diamond z) = (z \diamond 1)$ has the answer “yes”, since taking $x = 0, y = 1, z = 0$ we get $((0 \diamond 1) \diamond 1) \diamond (0 \diamond 0) = (0 \diamond 1)$, and evaluating both sides we get $1 = 1$, the equation holds. The assignment $x = 0, y = 0$ and $z = 1$ also works, but no others.

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Code number:

Question 9 (1 points, ☆): Let N be a countably infinite set. Associate with each $n \in N$ a regular language L_n . Prove or disprove the following statement: The infinite union $L_N = \bigcup_{n \in N} L_n$ is necessarily regular.

[illegible]

Code number:

Question 10 (1 points, ☆): Define the complexity class $\mathcal{NP}$ and demonstrate that it is closed under union and intersection. Include as much detail as possible in the proof. That $\mathcal{NP}$ is closed under union and intersection means that if the languages L_1 and L_2 both belong to $\mathcal{NP}$, then so does $L_1 \cup L_2$ and $L_1 \cap L_2$.

[illegible]

Collection of definitions

Theorem 1 (The regular pumping lemma). *Let L be a (usually infinite) regular language. Then there exists some positive integer n such that any $w \in L$ with $|w| \geq n$ can be decomposed as $w = xyz$, with $|xy| \leq n$ and $|y| \geq 1$, such that $w_i = xy^iz$, is also in L for all $i \geq 0$.*

Theorem 2 (The context-free pumping lemma). *Let L be an (infinite) context-free language. Then there exists some positive integer n such that any $w \in L$ with $|w| \geq n$ can be decomposed as $w = xuyvz$ with $|uyv| \leq n$ and $|uv| \geq 1$ such that $xu^iyv^iz \in L$ for all $i \in \mathbb{N}$ (including 0).*

Definition 1. A nondeterministic pushdown automaton (NPDA) is a septuple $M = (Q, \Sigma, \Gamma, \delta, q_0, z, F)$ where:

- Q is a finite set of states,
- Σ is the input alphabet,
- Γ is a finite set of symbols called the *stack alphabet*,
- $\delta : Q \times (\Sigma \cup \{\lambda\}) \times \Gamma \rightarrow 2^{Q \times \Gamma^*}$, where each set in the range is finite, is the transition function,
- $q_0 \in Q$ is the initial state,
- $z \in \Gamma$ is the *stack start symbol*,
- $F \subseteq Q$ is the set of final states.

Definition 2. A deterministic single-tape Turing machine M is defined by $M = (Q, \Sigma, \Gamma, \delta, q_0, \square, F)$, where

- Q is the set of states,
- Σ is the input alphabet,
- Γ is a finite set of symbols called the *tape alphabet*,
- δ is the transition function defined as $\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$.
- $\square \in \Gamma$ is a special symbol called the (tape) *blank*, we assume that $\square \notin \Sigma$,
- $q_0 \in Q$ is the initial state, and
- $F \subseteq Q$ is the set of final states.